

The Shape of the Universe as a Nested Topological Lattice: Prime Hierarchies and Cosmological Constant Quantization

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Abstract We present a nested topological lattice framework for the structure of the universe, grounded in the prime number hierarchy $\{37, 101, 1001\}$ and their associated repdigit sequences. The framework identifies three lattice levels — the 37-lattice (periodic), the 101-lattice (aperiodic/quasiperiodic), and the 1001-lattice (complete) — connected by phase-transition operators (1010, 101010, 111111) whose prime factorizations encode cross-domain physical constants. We anchor the framework to the recent result of Alexander, Bernardo, and Hui (Physical Review Letters, 2026) demonstrating topological quantization of the cosmological constant via the Chern-Simons-Kodama state, and to independent corroboration by Addazi et al. (2026) via pre-geometric gravity. We advance a falsifiable structural prediction: the cosmological constant occupies a topological sector whose index is constrained by the prime hierarchy $\{37, 101, 1001\}$, testable in principle via lattice quantum gravity simulations. We document a convergence of the integer 42 across four independent measurement contexts (magic cube constant, primary rainbow angle, Hubble constant under imperial unit conversion, and binary encoding of the phase-transition operator 101010) and propose this warrants investigation as a candidate topological invariant. We additionally document a verified biological frequency threshold at 101 kHz corresponding to the aperiodic bridge prime. The framework connects the Penrose quasicrystal paradigm, Viazovska's E8 lattice optimality, and Perelman's geometrization to a unified picture in which the universe is not a single manifold but a stack of prime-indexed topological lattices.

Keywords: cosmological constant, topological quantization, prime lattice, Chern-Simons-Kodama state, quasiperiodic structure, quantum gravity, nested topology, pre-geometric gravity, repunit sequences, 42 convergence

1. Introduction

The question of the universe's shape has driven mathematics and physics for two centuries. Gauss discovered intrinsic curvature but lacked field equations. Riemann built the tensor machinery but died before physics could use it. Poincaré posed the 3-manifold classification problem that required Perelman's Ricci flow to resolve. Einstein forced a static universe on dynamic equations. Each wall-hitter had the right tools for the wrong assumption.

The 2026 result of Alexander, Bernardo, and Hui [1] breaks a new wall. By demonstrating that the cosmological constant Λ is topologically protected in the Chern-Simons-Kodama (CSK) state of quantum gravity — analogous to quantized conductance in the quantum Hall effect — they establish that topology does not merely classify the universe's shape; it *protects physical values* from quantum fluctuations. This result reframes the cosmological constant problem from a fine-tuning question to a topological selection question.

Independent corroboration arrives from Addazi et al. [2], whose pre-geometric gravity (PGG) framework derives discrete topological sectors for Λ through spontaneous symmetry breaking of a fundamental gauge group, and from Hassfeld and Hebecker [3], whose Boundary Proposal provides an alternative to the Hartle-Hawking no-boundary

condition that survives rigorous quantum mechanical scrutiny.

This paper proposes that the shape of the universe is not a single manifold but a *nested stack of topological lattices*, each indexed by a prime number, connected by phase-transition operators whose mathematical structure encodes physical constants across multiple domains. We present the framework, document the evidence, advance a falsifiable prediction, and identify the open questions that bound the work.

2. The Nested Prime Lattice Framework

2.1 Overview

We define a nested prime lattice as a sequence of mathematical structures, each characterized by a prime number p , in which integer multiples of repunit strings (numbers of the form $111\dots1$) generate a periodic or quasiperiodic lattice. Transitions between lattice levels occur at specific break terms whose prime factorizations carry physical and mathematical significance.

2.2 The 37-Lattice (Level 1: Periodic)

The sequence $\{111, 222, 333, 444, 555, 666, 777, 888, 999\}$ forms the 37-lattice. Each term equals $k \times 111 = k \times 3 \times 37$ for $k = 1, \dots, 9$, with coefficients $\{3, 6, 9, 12, 15, 18, 21, 24, 27\}$ being multiples of 3. This lattice is periodic in the sense that all terms share a common prime factor structure.

The anchor prime 37 has documented mathematical significance as the first irregular prime [4] — the number at which Kummer's naive approach to Fermat's Last Theorem breaks down and modern algebraic number theory becomes necessary. This is not numerological; it is a structural threshold where simple periodicity fails and more complex algebraic architecture must be invoked. The 37-lattice is therefore interpreted as the periodic ground state.

2.3 Phase Transition I: The 1010 Operator

The break term $1010 = 2 \times 5 \times 101$ is not divisible by 37, marking the departure from the 37-lattice. Its factorization introduces the prime 101, which governs the next lattice level. Structurally, $1010 = 10 \times 101$.

2.4 The 101-Lattice (Level 2: Aperiodic/Quasiperiodic)

The sequence $\{1111, 2222, 3333, 4444, 5555, 6666, 7777, 8888, 9999\}$ forms the 101-lattice. Each term equals $k \times 1111 = k \times 11 \times 101$ for $k = 1, \dots, 9$, with coefficients $\{11, 22, 33, \dots, 99\}$ being multiples of 11. The structural analog is a quasiperiodic lattice: ordered but not periodic, in the sense of Penrose tilings [5] and the quasicrystalline phases discovered by Shechtman (Nobel Prize, 2011).

The anchor prime 101 is the 26th prime and a palindromic prime. It appears as atomic number 101 (Mendelevium) — the first synthetic element beyond the natural periodic table, marking another threshold between natural periodicity and human-constructed extension.

2.5 Phase Transition II: The 101010 Operator and the 42 Convergence

The break term $101010 = 2 \times 3 \times 5 \times 7 \times 13 \times 37$ is not divisible by 101, marking departure from the 101-lattice. Its most significant property is binary: 101010 in binary representation equals 42 in decimal ($42 = 32 + 8 + 2 = 2^5 + 2^3 + 2^1$). This is a mathematical fact, not an interpretation.

We document four independent contexts in which the integer 42 emerges, with no common causal mechanism identified:

Context	42 Emergence	Status
Mathematical	Magic constant of the 3×3×3 magic cube	Verified combinatorial fact
Physical — optics	Primary rainbow angle: ~42.3°	Verified; Descartes 1637

Physical — cosmology	$H_0 \sim 41.9$ mi/s/Mpc (Planck 2018 convention)	Unit-conversion artifact; noted with Hubble tension caveat
Computing	$101010_2 = 42_{10}$; ASCII 42 = * (wildcard)	Verified mathematical fact

Table 1. Four independent contexts of 42 emergence. The null hypothesis — that these are independent coincidences — is acknowledged. We propose this convergence warrants investigation as a candidate topological invariant.

Regarding the Hubble constant entry: current estimates range from 67.4 km/s/Mpc (Planck 2018 CMB) to 73.04 km/s/Mpc (SH0ES distance ladder), reflecting the unresolved Hubble tension. The imperial conversion $67.4 \text{ km/s/Mpc} \div 1.609 \text{ km/mi} \approx 41.9 \text{ mi/s/Mpc}$ approximates 42 under one convention only. This is noted as a unit-conversion correlation, not a physical prediction of the framework.

2.6 The 1001-Lattice (Level 3: Complete)

The terminal structure is governed by $1001 = 7 \times 11 \times 13$. The repunit $R_6 = 111111 = 3 \times 7 \times 11 \times 13 \times 37$ is the phase-transition operator to this level and serves as the complete prime assembly, containing all anchor primes from previous levels. The 1001-lattice is interpreted as the terminal level: the point of prime closure.

The 1001 structure carries a notable mathematical property: for any three-digit integer abc , $abc \times 1001 = abcabc$. This “number repetition” property means 1001 acts as a duplication operator — any pattern, multiplied by 1001, echoes itself.

2.7 The 5-Gap: A Structural Observation

Across all lattice levels, we observe a persistent 5-gap: no lattice level contains the digit 5 as an independently significant coefficient or anchor prime. In the 37-lattice, $555 = 5 \times 111$ is present but carries no structural role beyond its position as the 5th multiple. The 101-lattice has no 5-digit repdigit level. The anchor prime sequence $\{37, 101, 1001\}$ contains no prime with 5 as a factor. Whether this absence is a structural necessity of the lattice architecture or a selection artifact remains an open question. We document it without claiming physical significance.

3. Theoretical Anchors in the Literature

3.1 Topological Quantization of the Cosmological Constant

Alexander, Bernardo, and Hui [1] demonstrate that the cosmological constant Λ is topologically protected in the Chern-Simons-Kodama state of quantum gravity. The central result is:

$$\theta = 12\pi^2 / (\Lambda \blacksquare_{pI}^2) \bmod 2\pi$$

Because the CSK state must live in a particular θ -sector, Λ is constrained to discrete, topologically protected values — analogous to the quantized conductance in the quantum Hall effect. The authors state: the topology of spacetime renders quantum perturbations that should blow up the value of Λ inert, keeping the constant stable [1].

Addazi et al. [2] independently derive topological quantization of Λ through a distinct mechanism: pre-geometric gravity, in which spacetime itself emerges from spontaneous symmetry breaking of a gauge group $SO(1,4) \rightarrow SO(1,3)$ driven by a Higgs-like field. The cosmological constant is stabilized by entropic suppression of topological sector transitions, scaling as de Sitter entropy.

The convergence of two independent mechanisms on the same conclusion — Λ is topologically quantized — provides the theoretical foundation for interpreting the nested lattice hierarchy as a coordinate system for topological sector selection.

3.2 The Boundary Proposal and Universe Genesis

Hassfeld and Hebecker [3] propose an alternative to the Hartle-Hawking no-boundary condition. Rather than the shuttlecock geometry (which Turok et al. demonstrated fails rigorous quantum mechanical scrutiny), the Boundary

Proposal begins the universe with a dynamical spherical boundary — cutting a sphere at the equator, dropping half, and gluing the equator to expanding de Sitter space. A boundary is not a singularity; it is a phase transition. The proposal additionally admits flat torus universe creation from nothing if tensionless end-of-the-world branes from M-theory are available.

3.3 Geometric Foundations: Perelman, Viazovska, Penrose

Perelman's proof of Thurston's Geometrization Conjecture [6] establishes that every 3-manifold decomposes into pieces, each carrying one of eight geometric structures. This classifies 3-manifolds completely but does not identify which geometry the universe occupies. The nested lattice framework proposes that this selection is made by topological quantization rather than initial conditions.

Viazovska's proof of E_8 lattice optimality in dimension 8 [7] (Fields Medal, 2022) and Leech lattice optimality in dimension 24 demonstrate that specific lattice structures are globally optimal in specific dimensions. The nested prime lattice framework is consistent with this paradigm: each level occupies its prime-indexed dimension space at an optimal packing density.

Penrose's discovery of aperiodic tilings [5] and Shechtman's experimental observation of quasicrystals established a third state between periodic and random: quasiperiodic order. The 101-lattice is proposed as the cosmological analog — ordered but non-repeating, with a forbidden gap (the 5-gap) analogous to the quasicrystal's forbidden symmetry.

3.4 The 101 kHz Biological Threshold

Weinberg et al. [8] demonstrated in a prospective IRB-approved clinical trial that peripheral nerve stimulation probability under 0.4 T magnetic field pulses drops from 92.3% at 25 kHz to 7.7% at 101 kHz — a sharp threshold effect at the frequency corresponding in integer kilohertz to the aperiodic bridge prime. The wavelength of 101 kHz in rock ($\epsilon_r \approx 5$) is approximately 1.3 meters, matching the scale of fault zones and fracture networks. We note this correlation without claiming a causal mechanism between the lattice prime and the biological threshold.

4. The Period-2 Transition Hypothesis

We advance the following structural conjecture, explicitly labeled as a hypothesis requiring formal derivation:

Hypothesis (Period-2 Transition): The alternation between the integers 6 and 11 constitutes a Möbius-type transition operator between nested lattice levels. The structural basis: $6 = 2 \times 3$ (compactification primes, encoding dimensional reduction); $11 = \text{prime}$ (M-theory total dimensionality); $6 + 11 = 17$ (the heptadecagon, first constructible polygon beyond the pentagon, proven by Gauss at age 17); $2 \times (6 + 11) = 34$ (Möbius return, requiring two traversals for identity restoration, analogous to the 720° rotation symmetry of spin-1/2 particles). Two full 6-11 cycles are proposed to complete one lattice level transition.

This hypothesis is motivated by structural analogy with the CESR Möbius accelerator, in which particles require two full orbits to return to their betatron phase-space state. It requires formal derivation connecting the 6-11 recurrence to CSK topological phase transitions or pre-geometric gravity symmetry breaking. We present it as a conjecture motivating further investigation.

5. Falsifiable Prediction

The nested lattice framework, in conjunction with the CSK topological quantization result of Alexander et al. [1], yields the following structural prediction:

Prediction: The cosmological constant Λ occupies a topological θ -sector whose index is constrained by the prime hierarchy $\{37, 101, 1001\}$. If the Chern-Simons level k governing the CSK state is related to these primes, the allowed values of Λ form a specific discrete spectrum rather than a continuous range. This prediction is testable in principle via lattice quantum gravity simulations: if future simulations demonstrate

that Λ can take any continuous value within the CSK framework, the nested lattice constraint is falsified. If simulations show discrete sector selection governed by prime-structured level indices, the constraint survives.

We do not compute a numerical value for Λ . We predict a structural constraint on the allowed sector. This is consistent with the independent prediction of Addazi et al. [2] that Λ is discretized by symmetry breaking — our contribution is the proposed prime-number indexing of the discrete sectors.

6. Open Questions and Limitations

Open Question	Status
What physical process generates the prime hierarchy {37, 101, 1001}?	Unresolved. No mechanism derived.
Is the 5-gap a structural necessity or a selection artifact?	Unresolved. Documented as observation.
Does the period-2 (6-11) conjecture admit formal derivation?	Unresolved. Labeled as hypothesis.
Is the 42 convergence a topological invariant or coincidence?	Unresolved. Four independent contexts documented.
How does the nested lattice relate to E8 and Leech lattice optima?	Partial: structural consistency noted; no derivation.
What is the mechanism of the 101 kHz biological threshold?	Correlation noted; no causal mechanism proposed.

Table 2. Open questions bounded by this framework. Honest acknowledgment of gaps is a feature, not a weakness — it defines the research program.

7. Conclusions

We have presented a nested topological lattice framework as a structural model for the shape of the universe. The framework is grounded in verified mathematical properties of the prime hierarchy {37, 101, 1001}, anchored to peer-reviewed results in topological quantum gravity, and advances a falsifiable structural prediction testable via lattice quantum gravity simulations.

The central insight is that topology does not merely classify shape — it protects values. Just as the quantum Hall effect locks conductance against material imperfections, and just as Viazovska's E_8 lattice is globally optimal regardless of local perturbations, the universe's cosmological constant may be protected by the topological sector in which it resides. We propose that this sector is not arbitrary but is constrained by a prime-number hierarchy whose structure encodes the transition between periodic, quasiperiodic, and complete lattice levels.

The wall that shatters in this framework is the one between pattern and mechanism. We do not claim to have crossed it. We document where the wall stands, what lies on the other side, and what tools — lattice quantum gravity simulation, formal derivation of the period-2 conjecture, investigation of the 42 convergence — will be needed to cross it.

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This preprint has not undergone peer review. The anomaly correlation records (numerical sequence logs, field observation data) referenced as motivating material for this framework are maintained as internal Presignal documents and are explicitly excluded from this physics preprint.